

Adaptive Physics-Aware Raman Baseline Correction with Machine Learning Predicted Parameters

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ABSTRACT: Accurate baseline correction is critical for reliable Raman spectral interpretation. Traditional algorithmic methods often require manual tuning of regularization parameters, while recent machine learning and neural network approaches automate correction but lack generalizability and user control. We have developed a new approach to baseline correction which adaptively resolves baseline distortions without manual intervention - DIRAS (Dynamic Iterative Reweighted Autoregressive Spectral baseline correction). DIRAS uses a fixed regularization parameter (λ), which performs robust batch correction by iteratively reweighting residuals. We further used Structural Similarity Index Measure (SSIM) as an objective for λ optimization and trained a deep learning model to learn the nonlinear mapping between raw spectral features and optimal regularization. The resulting framework (DIRAS+) was capable of real-time spectrum-specific λ prediction. Applied to two SERS data sets, DIRAS+ outperformed ALS and SEALS in preserving peak fidelity, reducing intraclass variability and minimizing baseline distortion. Importantly, in downstream chemometric workflows, DIRAS improved calibration and model performance, yielding lower errors and enhancing analytical sensitivity. DIRAS and DIRAS+ together provide robust, scalable, and user-adaptable solutions for high-throughput Raman spectroscopy applications.



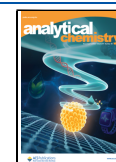
INTRODUCTION

Raman spectroscopy is a nondestructive analytical technique that leverages inelastic photon scattering to probe molecular vibrations, producing distinct spectral fingerprints for accurate chemical identification and structural characterization.^{1,2} Its versatility has led to widespread adoption in diverse areas such as biomedical diagnostics and environmental monitoring, underscoring its pivotal role in modern analytical chemistry.^{3–5} However, Raman spectroscopy faces technical challenges, notably the influence of both random and systematic noise on spectral measurements. High-frequency fluctuations can arise from detector noise, cosmic ray interference, or etaloning effects. A significant concern is baseline drift from sources such as fluorescence, which can obscure Raman features and complicate qualitative and quantitative analyses.⁶ Therefore, implementing accurate baseline correction is crucial to maintaining data integrity and ensuring reliable interpretations.

Existing baseline correction methods include polynomial fitting, which models the background using low-degree polynomials via least-squares. Due to their simplicity, such models often underfit complex or fluctuating baselines, whereas higher-order fits risk distorting genuine Raman peaks. Asymmetric Least Squares (ALS) and its variant Asymmetrically Reweighted Penalized Least Squares (arPLS) remain among the most widely adopted techniques. These methods iteratively adapt to spectra during fitting in an

attempt to preserve peaks, but they require careful tuning of smoothing (λ) and asymmetry parameters and frequently over smooth subtle spectral features.^{7–10} More advanced approaches, such as wavelet-based decomposition, attempt to isolate baseline drift by separating spectral content into distinct frequency components. However, their performance is sensitive to wavelet selection and decomposition depth, which lack standardized tuning protocols.^{11–13} Similarly, iterative strategies like the double sliding-window method (DSW) and its k-iterative variant (DSW^k) refine baseline estimates by progressively filtering spectral noise.¹⁴ Yet, these methods often conflate weak Raman signals with baseline variations, especially where signal-to-noise (SNR) is low, and rely heavily on manual parameter adjustment. This reliance on manual tuning undermines reproducibility and scalability. In high-throughput or real-time settings, small misconfigurations can lead to significant downstream errors such as misidentified peaks or distorted chemometric interpretations. For example, a

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recent study applied Raman spectroscopy to classify sugar types in fruit samples, but the baseline preprocessing introduced overfitting artifacts, resulting in negative-valued spectra that could compromise the accuracy of subsequent PLS-DA analysis.¹⁵

Recent efforts have integrated deep learning frameworks, including ResNet, U-Net, and convolutional autoencoders, into the baseline correction process. These methods automate the correction process by simultaneously denoising and removing baselines, thereby preserving essential spectral features without explicit parameter tuning. Additionally, methods such as S-ModPoly have introduced adaptive segmentation strategies using localized polynomial fits, significantly enhancing performance in high-throughput environments.^{16–18} However, deep learning-based approaches remain limited by substantial data requirements, fixed deterministic baselines, high computational demands, and a lack of interpretability and user control. Many existing models are trained on narrow or idealized spectral conditions, making them poorly equipped to handle the signal-to-noise variability, spectral complexity, and batch-level heterogeneity typical of real-world Raman data sets. For example, a recent report by Cui et al. presented an automated airPLS parameter optimization scheme using machine learning.¹⁹ While this removes the need for manual tuning, the workflow remains constrained by airPLS's limitations, notably its susceptibility to failure in cases of strong fluorescence backgrounds or low signal-to-noise ratios. The authors also acknowledge their model's applicability is largely bounded by the spectral conditions represented in the training set, raising concerns about generalization to noisy or heterogeneous data sets. This inflexibility is a critical limitation in preprocessing Raman data, where the baseline structure is often ambiguous and ground truth is undefined. Without expert oversight or flexibility, such rigid outputs risk compromising scientific interpretability. Moreover, few models are evaluated for their ability to generalize across batches or data sets, raising questions about their scalability and reliability.^{18,20} These gaps highlight the need for baseline correction strategies that are not only accurate but also adaptive, explainable, and suited to deployment in high-throughput or semisupervised pipelines.

Scaling computational resources or relying on conventional mathematical models alone does not solve baseline-correction challenges. Instead, there is a clear need for an integrated framework that automates baseline estimation while retaining interpretability by explicitly modeling baseline features. Such an approach would minimize downstream analytical errors, enable expert oversight where needed, and provide the flexibility required for diverse Raman applications without necessitating retraining or extensive parameter tuning. A notable step toward more expressive mathematical modeling is the SEALS algorithm,²¹ which combines autoregressive spectral estimation with adaptive reweighting to enhance baseline correction. At its core, SEALS employs a spectral estimation kernel (SEkel), derived via inverse Fourier transform and autoregressive modeling, to capture the energy distribution of the residuals and adjust smoothing weights dynamically. This enables better preservation of spectral structure near broad or high-intensity peaks, where traditional methods often distort the baseline. Control parameters λ , ω and ζ modulate the aggressiveness of smoothing, allowing sharper adaptation in low-signal or noisy regions. However, SEALS relies on fixed parameter settings, making it less

responsive to shifts in baseline complexity or SNR. As a result, SEALS may overcorrect or undercorrect depending on the data set, limiting its applicability in variable experimental contexts.

Addressing the limitations of fixed-parameter approaches, we developed the Dynamic Iterative Reweighted Autoregressive Spectral (DIRAS) baseline correction method, which leverages dynamic, data-driven parameter adjustments to enhance both robustness and generalizability. DIRAS delivers significantly improved consistency and accuracy compared to SEALS and conventional least-squares methods. Building on this foundation, DIRAS+ is a hybrid extension that incorporates a custom CNN encoder with an XGBoost regressor to predict optimal correction parameters automatically. This enhancement substantially boosts correction accuracy across diverse spectral conditions while still allowing user-driven parameter adjustments for edge cases - an important capability given the lack of a universally accepted baseline standard in Raman spectroscopy. Moreover, DIRAS+ streamlines batch-level correction, which is essential for applications such as sensor calibration and spectral mapping. When applied to surface-enhanced Raman spectroscopy (SERS) sensors, DIRAS+ demonstrates markedly better analytical accuracy than traditional SEALS and ALS-based methods.

METHODS

Reference Baseline Models. All baseline correction algorithms employ a common backbone (eq S1) but differ in how weights are defined and updated. Asymmetric Least Squares (ALS) penalizes positive residuals more heavily than negative residuals to suppress peak influence (eq S4). Asymmetrically Reweighted Penalized Least Squares (arPLS) adapts weights iteratively based on the statistical distribution of residuals, reducing large-peak bias while preserving baseline fit in low-signal regions (eq S5). Spectral Estimation-based Asymmetrically Reweighted Penalized Least Squares (SEALS) further refines arPLS by incorporating a spectral estimation kernel derived from inverse Fourier energy analysis to modulate weights according to both residual magnitude and spectral energy (eqs S6–S8). All parameter values and convergence criteria (relative change $<10^{-6}$) are detailed in the Supporting Information.

Dynamic Iterative Reweighted Autoregressive Spectral Baseline (DIRAS). DIRAS builds on SEALS by integrating autoregressive spectral modeling with dynamically evolving weight controls to achieve robust baseline correction (eqs S9–S15). In each iteration, residuals are reweighted using adaptive logistic and spectral-kernel functions, an Autoregressive model is refitted to update the spectral energy kernel, and a new baseline is obtained via weighted PLS (Penalized Least squares). The procedure initializes with a baseline-agnostic kernel and repeats until relative changes in both baseline and weights fall below 10^{-6} . Full algorithmic details and derivations are provided in the Supporting Information Methods section.

Simulated Data Set. To train and evaluate the model we constructed a synthetic data set that is representative of the experimental variations in signal and noise that we experience in experimental spectra. Each spectrum was defined on a uniform wavenumber axis from 500 cm^{-1} to 1800 cm^{-1} , with a resolution of 1,015 equally spaced points to simulate typical data acquisition conditions. The spectral features were generated by adding a variable number of peaks, sampled from a uniform distribution between 5 and 30, to each

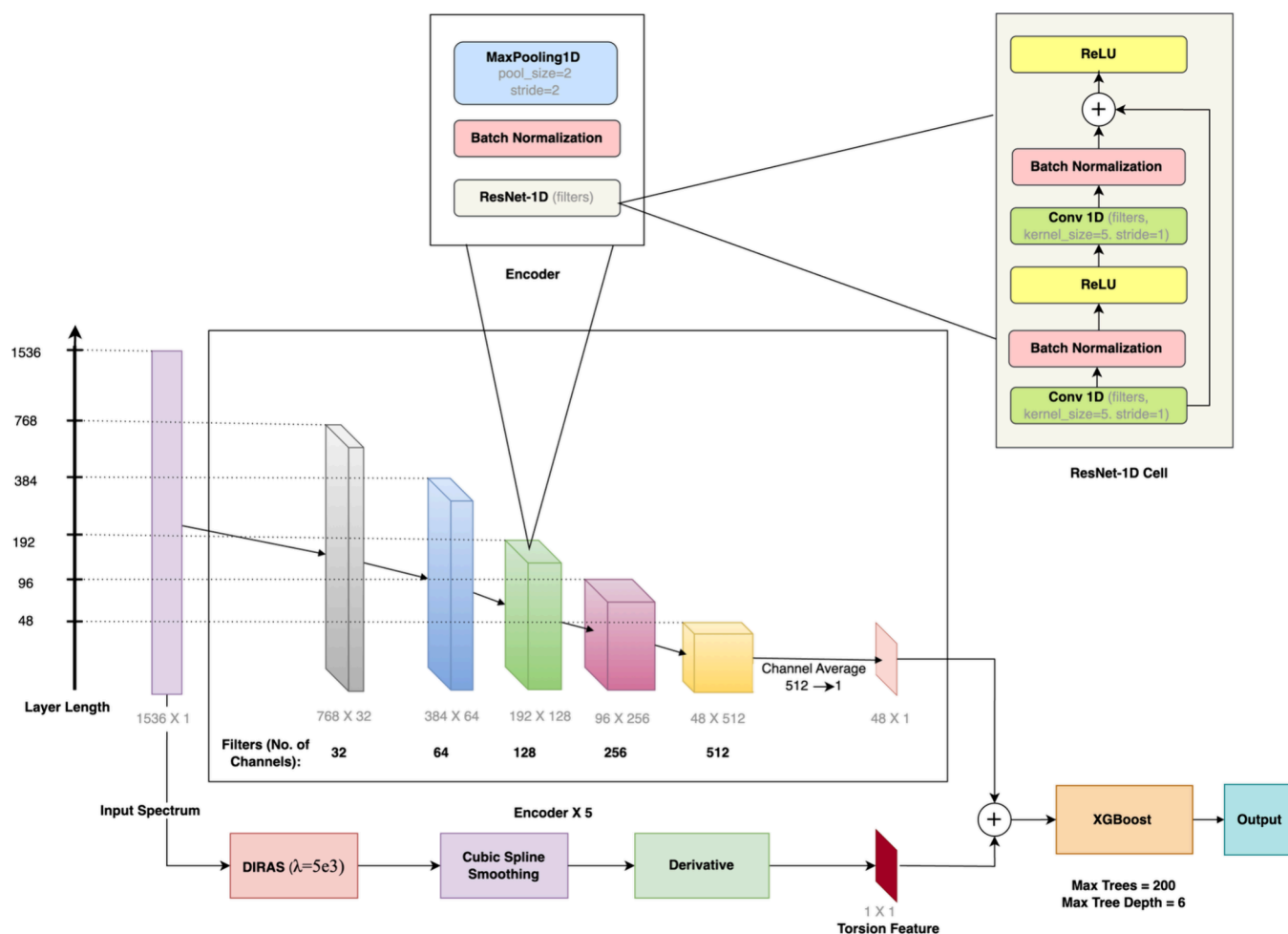


Figure 1. Architecture of the torsion-augmented ResNet1D regression model.

spectrum. These peaks were modeled using Voigt profiles, with their center positions, widths ($5\text{--}30\text{ cm}^{-1}$), and intensities ($0\text{--}100\text{ au}$) all sampled from uniform distributions across their respective ranges. To emulate common experimental artifacts, complex baselines were introduced. A sinusoidal model was used to simulate instrumental drift and fluorescence backgrounds, with amplitudes and frequencies randomized within experimentally determined ranges (3×10^{-4} to $8 \times 10^{-4}\text{ cm}^{-1}$). Following the addition of the baseline, Gaussian-distributed noise ($\sigma = 0.3$) was introduced to simulate measurement noise. Finally, to ensure robustness across different signal conditions, the Signal-to-Noise Ratio (SNR) was systematically varied. This was achieved by scaling the combined signal and baseline before the addition of the constant Gaussian noise. For large-scale data set generation, this scaling factor was sampled uniformly between 0.25 and 1.0, creating a diverse data set with effective SNR values, based on peak amplitude, spanning approximately 3 to 13.

Evaluation Metrics. To quantitatively evaluate baseline correction performance, we employed a combination of error-based and structural similarity metrics. Specifically, we used Mean Squared Error (MSE), Mean Absolute Error (MAE), and the Structural Similarity Index Measure (SSIM). MSE captures the average of squared deviations between the estimated baseline $\hat{z}$ and the ground truth baseline z , while MAE reflects the average absolute difference. To assess structural fidelity, we adapt the SSIM framework which was

originally developed for images to 1D spectral baselines by treating the mean, variance, and covariance of the two vectors analogously to luminance, contrast, and structure.²²

$$\text{SSIM}(\hat{z}, z) = \frac{(2\mu_{\hat{z}}\mu_z + C_1)(2\sigma_{\hat{z}z} + C_2)}{(\mu_{\hat{z}}^2 + \mu_z^2 + C_1)(\sigma_{\hat{z}}^2 + \sigma_z^2 + C_2)} \quad (1)$$

where $\mu_{\hat{z}}$ and μ_z are the means, $\sigma_{\hat{z}}^2$ and σ_z^2 the variances, and $\sigma_{\hat{z}z}$ the covariance between $\hat{z}$ and z . C_1 and C_2 are small stability constants. Since SSIM values range from 0 to 1, with higher values indicating greater similarity, we report error as 1-SSIM. Hyperparameter optimization for all models was conducted using either Optuna²³ (a framework for Bayesian hyperparameter optimization) or grid search. The objective function was defined as a weighted combination of MSE, MAE, and 1-SSIM, enabling consistent model evaluation across varying spectral complexities.

Model Architecture and Training. A total of 22,000 synthetic Raman spectra were generated by randomizing peak positions, amplitudes, and widths, while superimposing baseline distortions of varying amplitude and frequency. For each synthetic spectrum, the optimal regularization parameter (λ) for baseline correction was identified via Optuna hyperparameter optimization over 300 trials, minimizing a composite 1-SSIM-based loss function. Each spectrum was standardized to 1536 points via zero-padding or truncation prior to training. A ResNet-inspired 1D autoencoder was

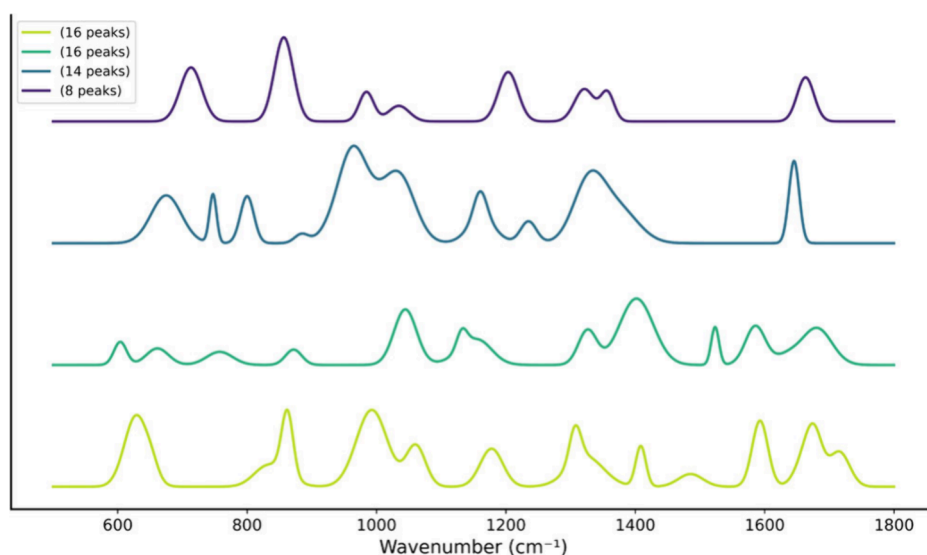


Figure 2. Synthetic Raman spectra illustrating increasing structural complexity. The plot displays spectra with 8 (lime), 14 (teal), and 16 peaks (blue and violet). The spectra are vertically offset for clarity, and the two 16-peak spectra represent different peak configurations.

employed to compress spectra into a 48-dimensional latent space through five encoder blocks. As illustrated in Figure 1, each encoder block comprised two Conv1D layers (kernel = 5, stride = 1) with an increasing number of filters (32, 64, 128, 256, 512), each followed by Batch Normalization and ReLU activation, and a MaxPooling1D layer (pool size = 2, stride = 2). The decoder mirrored this structure with UpSampling layers. The model was trained to reconstruct the input spectra using the Adam optimizer and the 1-SSIM loss function, preserving spectral structure and encouraging denoising. Independently, torsion features were computed by applying DIRAS with a low, fixed smoothing parameter, followed by cubic spline smoothing, second derivative estimation, and integration of the absolute curvature (scaled by 10^3). The resulting torsion values were concatenated with the 48-dimensional latent features, forming a 49-dimensional composite representation. This combined feature vector was input to an XGBoost²⁴ regressor configured with 200 trees and a maximum tree depth of 6, along with 80% row and column subsampling. The regressor was trained to predict $\log_{10}(\lambda)$ using an MSE loss function.

SERS Substrate Surface Modification. Gold-coated Tentagel particles (AuTG) were synthesized by modifying NH_2 -functionalized Tentagel beads (20 μm , Rapp Polymere GmbH, M30202) with 150 nm gold nanoparticles (AuNPs, Sigma-Aldrich, 742058). One mL suspension of AuNPs was incubated overnight at 4 $^\circ\text{C}$ with 0.3 mg/mL Tentagel beads to yield AuTG.²⁵ These were subsequently washed, pelleted by centrifugation, and functionalized with 100 mM 3-mercaptopropionic acid (3-MPA, Sigma-Aldrich, M5801) in phosphate-buffered saline (PBS). Functionalization was performed at room temperature for 12 h to produce AuTG-MPA. AuTG-MPA was functionalized with 2,6-diaminoanthraquinone (DAAQ) (Sigma-Aldrich, D122007) to obtain AuTG-MPA-DAAQ. Briefly, AuTG-MPA particles were incubated for 30 min at room temperature in a freshly prepared solution containing 100 mM 1-ethyl-3-(3-(dimethylamino)propyl) carbodiimide (EDC) (Sigma-Aldrich, 39391) and 50 mM *N*-hydroxy succinimide (NHS) (Thermo Scientific, 24510) in dH_2O at a 1:1 volume ratio with the AuTG-MPA suspension.²⁶ A 5 mg/mL solution of DAAQ in 99% ethanol

was then added to the carbodiimide-activated AuTG-MPA at a 1:2 volume ratio (activated AuTG-MPA suspension) and incubated for 3 h at room temperature. Following incubation, AuTG-MPA-DAAQ particles were washed thoroughly with PBS. For pH studies, AuTG-MPA particles were suspended in buffered PBS (phosphate buffered saline) at pH values ranging from 3 to 8, in 1-unit intervals. For each pH condition, ~ 100 SERS spectra were acquired from all 3 beads.

Raman Spectroscopy. SERS spectra were acquired using a Renishaw InVia Raman microscope equipped with a 785 nm laser, a 1200 grooves/mm grating, and a 50 $\times$ objective lens (Olympus LUMPlanFL). The system was calibrated daily using the 520 cm^{-1} Raman band of silica. Each spectrum was recorded using a 1-s integration under a laser power of 10 mW. Spectral acquisition was performed using a point-mapping approach on individual beads. For each experimental run, a spectral map was collected from at least three independent particles, resulting in 296 spectra for AuTG-MPA and 419 spectra for AuTG-MPA-DAAQ. Samples were prepared by placing 20 μL of bead suspension onto a CaF_2 window, followed by coverslip application to ensure uniform sample thickness and minimize optical distortion.

Computation Hardware and Processing. All simulations and data preprocessing were conducted in Python (v3.10). Consequently, the model was trained on a Google Colab environment running a Jupyter Notebook with a Python (v3.10 kernel). The models were implemented in TensorFlow (v2.), and statistical evaluations were performed using SciPy and scikit-learn libraries. DIRAS+ was trained and tested on a cloud system equipped with an Nvidia A100 GPU, 12 CPUs, and 80 GBs of GPU memory. All inferences were run on a general-purpose laptop running Python (v3.10).

RESULTS AND DISCUSSION

DIRAS Achieves Highest Fidelity with Minimal λ Sensitivity. Dynamic Iterative Reweighted Autoregressive Spectral Baseline Correction (DIRAS) is an approach designed to reduce sensitivity to the smoothing parameter λ , which governs the trade-off between signal fidelity and baseline smoothness. To evaluate this, we compared DIRAS to three

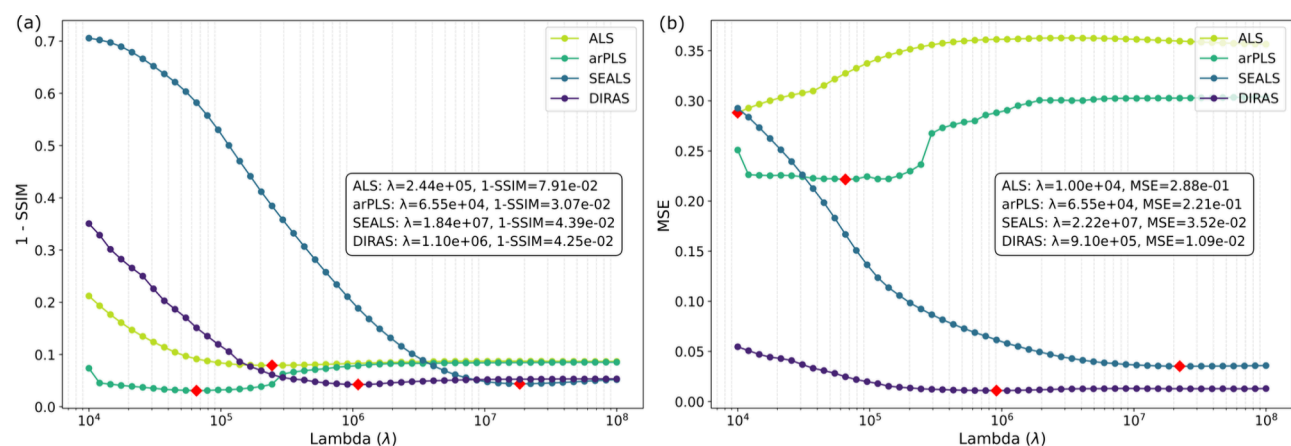


Figure 3. Regularization parameter (λ) sensitivity analysis under low signal-to-noise ratio (SNR) conditions. (a) 1-SSIM and (b) MSE values plotted as a function of λ (10^4 - 10^8) for ALS (lime), arPLS (teal), SEALS (blue), and DIRAS (violet). Filled red diamond markers indicate the λ corresponding to the best performance metric for each method.

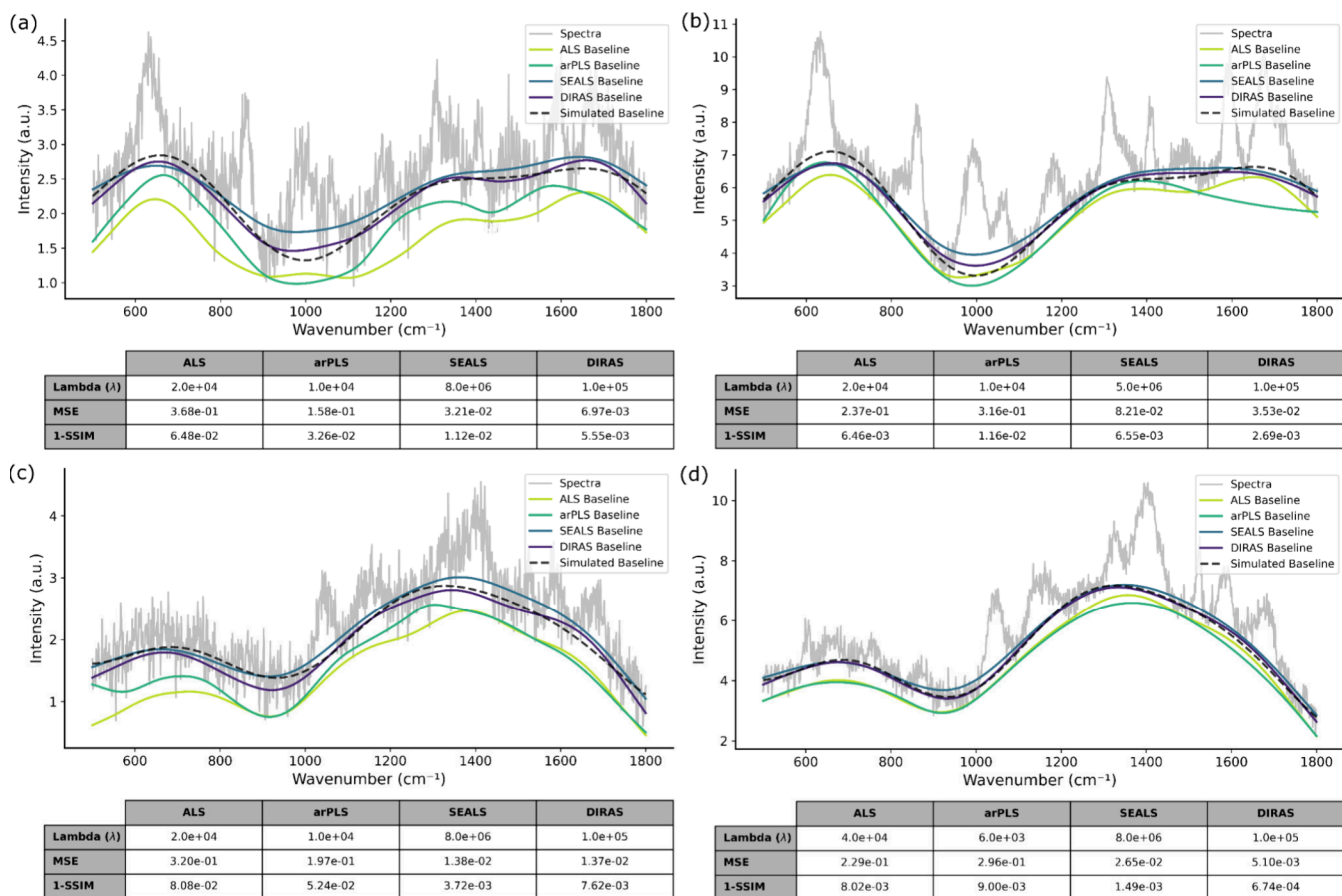


Figure 4. Baseline correction performance across varying SNR levels for two distinct synthetic spectra. Panels (a) and (b) depict the same underlying spectrum with fixed baseline shape, and panels (c) and (d) represent a different spectrum and baseline profile, at two SNR levels. Raw spectrum (gray) is overlaid with baselines estimated using ALS (lime), arPLS (teal), SEALS (blue), and DIRAS (violet). The ground truth baseline is shown as a black dashed line. Tables beneath each subplot provide the λ values, MSE, and 1-SSIM values corresponding to each method.

widely used baseline-correction algorithms: Asymmetric Least Squares (ALS), Adaptive Reweighted Penalized Least Squares

(arPLS), and Spectral Estimation-based Asymmetrically Reweighted Penalized Least Squares (SEALS). The compar-

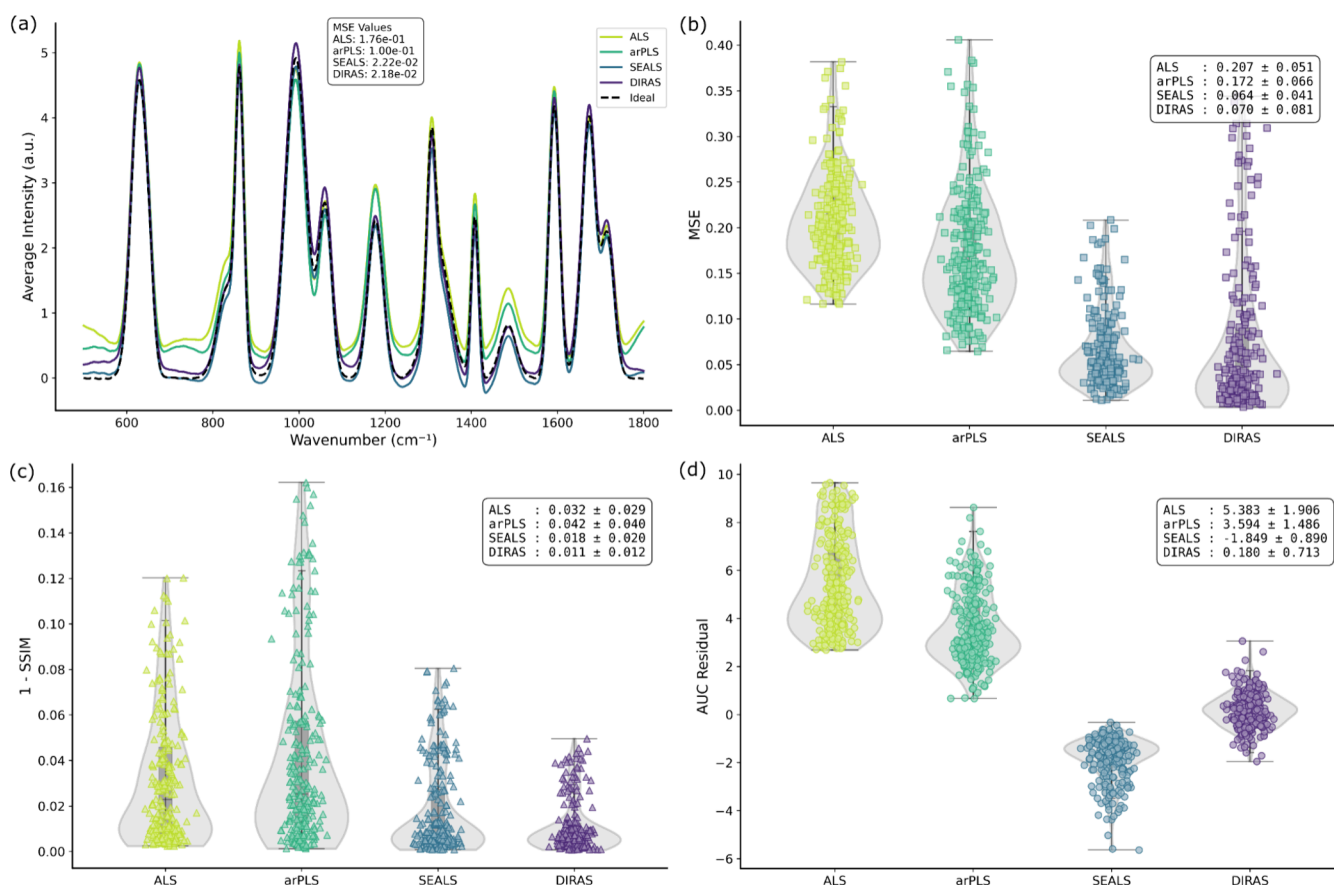


Figure 5. Baseline correction performance across a synthetic spectral batch ($n = 300$). (a) Average normalized spectra following baseline correction using ALS (lime), arPLS (teal), SEALS (blue), and DIRAS (violet). Violin plots showing distribution of (b) MSE, (c) 1-SSIM, and (d) AUC residual for each method. Each point represents a single spectrum from the batch.

ison was performed on a comprehensive data set of synthetic Raman spectra, which we generated to exhibit varied baseline profiles, peak complexities (Figure 2), and noise levels (Figure S1) similar to those experienced in experimental spectra.

To rigorously evaluate performance and λ -sensitivity, we conducted a 50-point logarithmic grid sweep of λ from 10^4 to 10^8 for the four baseline correction methods. Two orthogonal performance metrics were used: Mean Squared Error (MSE) for pointwise accuracy and 1-SSIM for structural fidelity (Figure 3). For low-SNR spectra, ALS and arPLS exhibited persistently high MSE values (0.288 and 0.221, respectively) with steep response curves, indicating pronounced instability, although their 1-SSIM minima (0.0791 and 0.0307) were modestly low. In contrast, SEALS and DIRAS achieved substantially lower errors. SEALS attained a minimum MSE of 0.0352 but showed strong λ -dependence, as evidenced by a steep performance slope. DIRAS, by comparison, yielded the lowest minimum MSE (0.0109, $\sim 3.2\times$ lower than SEALS) together with a competitive 1-SSIM (0.0425), while maintaining the flattest slope at the optimum λ . This robustness indicates that DIRAS preserves both numerical and structural fidelity across a broad λ range, eliminating the need for precise parameter tuning - an especially valuable property in experimental settings where ground truth baselines are not

available. Similar trend was observed for higher SNR spectra (Figure S2).

DIRAS Provides More Consistent Baseline Subtraction than ALS, arPLS, or SEALS. We evaluated each algorithm's baseline fitting accuracy using synthetic spectra with known ground truth baselines. To ensure a fair comparison, key regularization parameters were manually selected. Across all tested conditions, both ALS and arPLS consistently underfitted the baseline, struggling to capture its more complex features. In contrast, SEALS and DIRAS followed the baseline curvature more accurately. However, at low SNR conditions, SEALS exhibited significant overfitting (Figures 4a, 4c). In comparison, DIRAS successfully avoided both underfitting and overfitting, decisively outperforming all other methods when evaluated by MSE and 1-SSIM. While all methods improved at higher SNR (Figures 4b, 4d), DIRAS maintained its clear superiority. This trend was consistent for every spectrum tested (Figure S3). Crucially, DIRAS achieved this superior performance across all SNR conditions without needing any change to its λ parameter. This highlights its robustness and confirms the results of our lambda sensitivity experiment (Figure 3).

We next tested each method's robustness on large-scale data sets representative of mapping or time-course experiments. For this, we generated four variants of synthetic spectra each

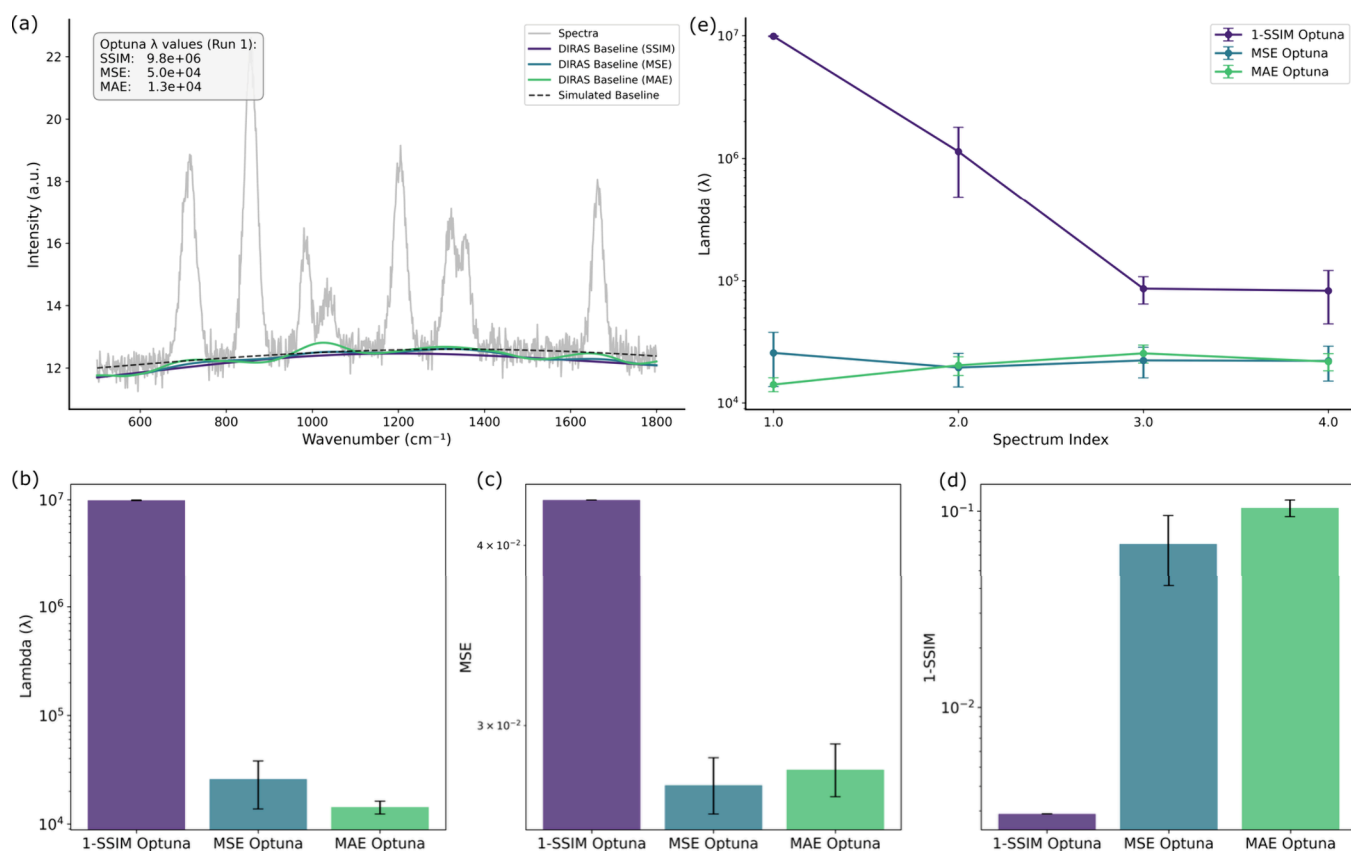


Figure 6. Comparison of Optuna-optimized regularization using three different loss functions. (a) Representative baseline fits for Synthetic Spectrum 1. Solid lines show DIRAS baselines obtained by minimizing 1-SSIM (violet), MSE (blue), and MAE (teal); the ground truth baseline is dashed black. (b–d) Bar plots (mean $\pm$ SD, $N = 20$ Optuna trials) of the (b) optimized λ values, resulting (c) MSE and (d) 1-SSIM for Spectrum 1. (e) Mean $\pm$ SD of optimized λ values across four synthetic spectra of increasing baseline complexity.

consisting of 300 samples with randomly varying baseline shapes and SNR levels and applied batch correction using a fixed, empirically chosen λ for each algorithm. Because fine-tuning parameters for hundreds of spectra is impractical, this setup mimics real workflows where a single λ must serve diverse baseline profiles. By comparing corrected outputs against the known ground truth, we quantified small fitting errors that could accumulate and potentially mislead downstream analyses.

The overfitting tendency of SEALS and the underfitting behavior of arPLS and ALS already seen in Figure 4 become much more pronounced in the batch correction analysis, as shown in Figure 5. For the first synthetic spectrum batch ($n = 300$), ALS and arPLS performed poorly, with average MSE values of 0.207 ± 0.051 and 0.172 ± 0.066 , respectively, and high AUC residual values suggesting underfitting (ALS: 5.383 ± 1.906 ; arPLS: 3.594 ± 1.486). SEALS achieved the lowest MSE (0.064 ± 0.041), but this was accompanied by signs of overfitting, as indicated by a negative AUC residual (-1.849 ± 0.089) and relatively high 1-SSIM (0.018 ± 0.020). In comparison, DIRAS obtained a slightly higher MSE (0.070 ± 0.081) but achieved the lowest 1-SSIM (0.011 ± 0.012) and a near-zero AUC residual (0.180 ± 0.713), indicating accurate baseline correction without structural distortion. Similar trends were observed in the remaining three spectra (Figure S4 a–i), DIRAS consistently achieved the best performance and structurally accurate baselines.

Classical baseline-correction methods for Raman spectra such as ALS and its variants airPLS, arPLS require careful, trial-

and-error tuning of the smoothing parameter λ to balance background removal against peak preservation.¹⁸ In practice, this ad hoc adjustment is time-consuming and often fails to generalize: under penalizing (high λ) leads to underfitting, leaving residual low-frequency drift, while over penalizing (low λ) can overfit and flatten genuine peaks, especially in low-SNR data.²⁰ Moreover, these methods are highly sensitive to spectral noise and variability, i.e. noise spikes may be misinterpreted as signal, causing baseline wandering so that parameters tuned on one data set frequently break down on another. Even SEALS, which introduces edge-aware smoothing, can overfit broad features when λ is not at its optimum (e.g., Figure S3a).^{27–29} This lack of transferability undermines reproducibility: analysts often must reoptimize parameters for each data set, yielding baseline corrections that are not directly comparable.

By introducing residual-magnitude dependent control terms alongside a fixed global smoothness penalty, DIRAS gracefully balances drift suppression and local feature preservation. Instead of relying on a single λ to both flatten slow background drift and ignore sharp spectral features, it uses dynamic weights derived from an autoregressive spectral estimator and simple logistic functions of the residuals to attenuate points whose deviation exceeds the ambient noise level. Noise-scale fluctuations remain governed by the baseline's nominal stiffness, while genuine peaks and troughs progressively override smoothing at high-energy locations or “lock in” true baseline points. This adaptive strategy greatly reduces sensitivity to λ for peak preservation, yet retains a global

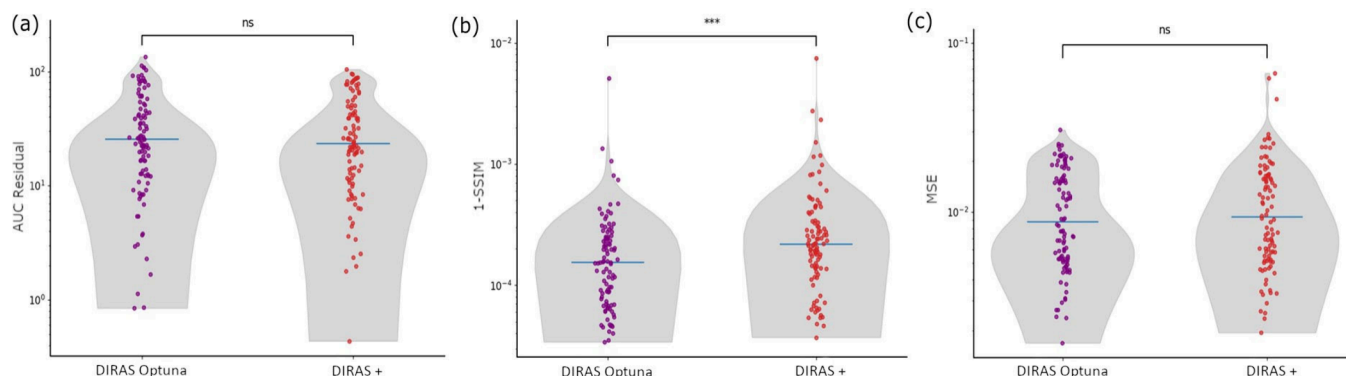


Figure 7. Comparative performance of DIRAS baselining with Optuna-optimized λ (purple) versus DIRAS+-predicted λ (red) across 100 synthetic test spectra: violin plots of (a) AUC residual, (b) 1-SSIM, and (c) mean squared error (MSE). Horizontal lines indicate group means. Statistical significance was assessed by paired t tests (** $p < 0.001$; ns = not significant).

curvature penalty for drift suppression that may still require tuning. Separate heuristics for positive versus negative residuals, weak-peak masking, and convergence thresholds ensure smooth, stable iterations without hard cut-offs. As a result, DIRAS delivers consistent baseline estimates across diverse spectroscopic contexts, simplifies peak-related parameter selection, and remains practical for large data sets with varying baseline characteristics.

Despite the robustness of DIRAS across a wide range of baseline drift and noise characteristics for a given λ , its performance can be further improved when λ is set within the optimal range, yielding a more accurate fit. While manual λ selection is appropriate for small numbers of spectra analyzed by experienced users, mapping and time-course experiments generate 1000s of spectra, often with very different baseline profiles and would benefit from an automated λ selection and baseline processing. To meet this need, we expanded upon DIRAS to develop the advanced automated baseline-correction framework DIRAS+. We aimed to build a machine-learning model capable of predicting optimal parameter values while preserving user flexibility for manual adjustments. However, training such a model necessitates high-quality ground truth λ values. First, we therefore evaluated MSE, MAE, and 1-SSIM as loss functions across four synthetic spectra with increasing baseline complexity to guide λ optimization. This allowed us to assess which objective best aligned with human-perceived fit quality and structural accuracy.

SSIM-Based λ Selection Outperforms MSE/MAE for True Baseline Recovery. To refine the DIRAS regularization parameter (λ), we conducted 20 independent Optuna optimizations for each spectrum, each running for 200 trials. A representative example is shown for a spectrum with a simple baseline, which logically requires a high λ value for a smooth fit (Spectrum 1, Run 1; Figure 6a). In this case, optimizations using MSE and MAE loss function (MSE-opt, MAE-opt) converged on low λ values (5.0×10^4 and 1.3×10^4 respectively), resulting in baselines that overfit high-frequency noise and distorted the true baseline shape. In contrast, the optimization using a 1-SSIM loss function (1-SSIM-opt) consistently selected a substantially higher λ of 9.8×10^6 , yielding a smoother, visually coherent fit. Across all 20 runs for Spectrum 1, 1-SSIM-opt averaged ($9.88 \pm 0.06 \times 10^6$) nearly 2 orders of magnitude above MSE-opt ($2.59 \pm 1.21 \times 10^4$) and MAE-opt ($1.43 \pm 0.19 \times 10^4$) (Figure 6b). 1-SSIM-opt repeatedly selected a smoother baseline that, despite a slight

increase in MSE ($4.30 \pm 0.01 \times 10^{-2}$), delivered a markedly lower structural error (1-SSIM = $2.90 \pm 0.001 \times 10^{-3}$), avoiding the overfitting of MSE-opt and MAE-opt (Figures 6c and 6d). Importantly, despite variability in λ , DIRAS's performance metrics remained tightly clustered, confirming that it does not require precise tuning to deliver reliable correction (Figure 6b-d). Moreover, only 1-SSIM-opt adapted to increasing baseline complexity: as spectra progressed from simple (Spectrum 1, $\lambda \approx 9.98 \times 10^6$) to complex (Spectrum 4, $\lambda \approx 1.1 \times 10^5$), 1-SSIM-opt progressively decreased λ , whereas MSE-opt and MAE-opt remained effectively static around 10^4 (Figure 6e).

In addition to evaluating individual metrics, we also explored normalized, weighted composite approaches that combined 1-SSIM with either MSE or MAE. However, in every case the inclusion of MSE or MAE biased the selection toward less reliable λ values, reflecting their greater susceptibility to numerical error and overfitting. Rather than enhancing robustness, these composites diluted the discriminative strength of 1-SSIM. Consequently, we retained 1-SSIM alone as the optimal selection criterion, as it consistently produced more reproducible and experimentally logical outcomes. Additional examples for Spectra 2, 3, and 4 are provided in Figures S5 and Table S1.

DIRAS+ Automates λ Selection Matching Optuna Performance. Building on the ground truth defined by 1-SSIM loss, we developed DIRAS+, a machine learning extension of the DIRAS framework that automates λ selection. DIRAS+ predicts λ directly from spectral and curvature-derived features, enabling rapid, context-specific baseline correction without manual parameter tuning. A custom hybrid model (CNN + XGBoost) was trained to predict $\log_{10}(\lambda)$. On a hold-out validation set, the model achieved a mean squared error of 0.0202 $\log_{10}$ -units (RMSE ≈ 0.14) (Figure S6). Such deviations lie around the optimum λ of the DIRAS cost surface, where multiple λ values produce indistinguishable baseline reconstructions. This behavior is consistent with the flattest error landscape observed for DIRAS in Figure 3, confirming that the predicted λ values remain well within the range where baseline reconstructions are unaffected. Together, these results demonstrate that DIRAS+ learns a robust mapping from spectral features to optimal correction parameters and achieves Optuna-level accuracy without the need for per-spectrum tuning.

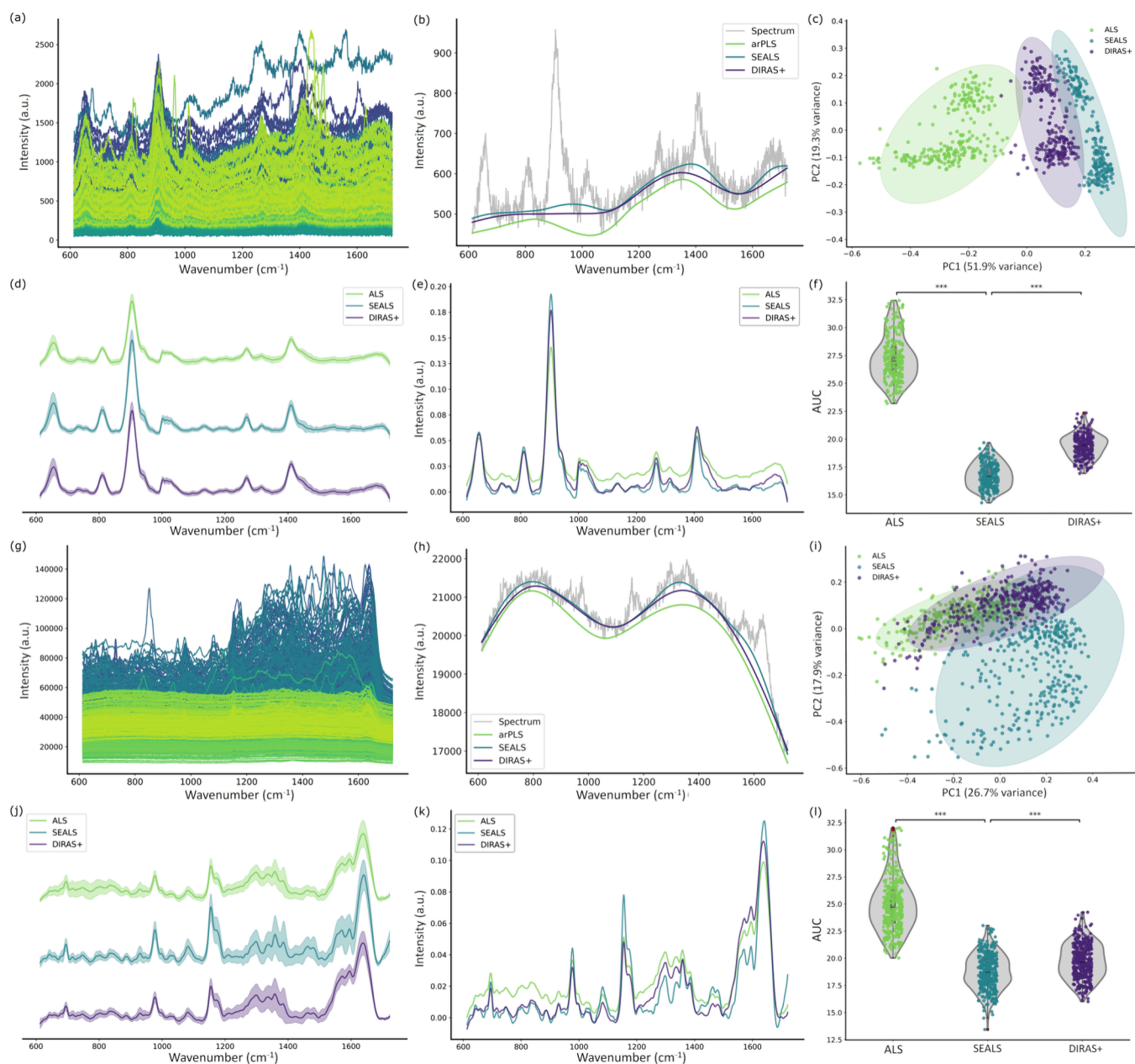


Figure 8. Baseline-correction performance on two experimental SERS data sets AuTG-MPA ($n = 296$; panels a–f) and AuTG-MPA-DAAQ ($n = 419$; panels g–l) for three methods: ALS (teal), SEALS (blue), and DIRAS+ (violet). (a, g) All raw spectra of SERS data set. (b, h) Representative baseline fits overlaid on the original spectra (gray). (c, i) PCA score plots (PC1 vs PC2) of the corrected spectra. (d, j) Mean corrected spectra, normalized and offset for clarity. (e, k) Overlaid average spectra for each method. (f, l) Violin plots of the AUC for each method; horizontal lines denote group means, and unpaired ANOVA reveals significant differences across methods ($***p < 0.001$).

To evaluate the generalizability of DIRAS+, we applied it to a test set of 100 synthetic Raman spectra featuring entirely novel baseline and noise profiles. For each spectrum, DIRAS+'s internal model predicted and applied an optimal regularization parameter λ , while a parallel Optuna-based search (minimizing 1-SSIM) identified the parameter yielding the best achievable correction. As shown in Figure 7a–c, DIRAS+ and Optuna produced statistically indistinguishable results across all metrics. DIRAS+ achieved an average MSE of $(1.26 \pm 1.07) \times 10^{-2}$, a 1-SSIM of $(4.00 \pm 8.00) \times 10^{-4}$, and an AUC residual of $(3.39 \pm 2.76) \times 10^1$. Optuna yielded $(1.12 \pm 0.70) \times 10^{-2}$ MSE, $(2.00 \pm 5.00) \times 10^{-4}$ 1-SSIM, and $(3.60 \pm 3.04) \times 10^1$ AUC. Paired t tests confirmed no significant difference in MSE ($p = 6.19 \times 10^{-2}$) or AUC ($p = 2.62 \times$

10^{-1}). Although the 1-SSIM difference was statistically significant ($p = 2.11 \times 10^{-5}$), both methods remained within the 10^{-4} range, indicating negligible practical disparity in structural fidelity. These results demonstrate that DIRAS+ delivers baseline-correction performance equivalent to spectrum-specific Optuna tuning, yet without per-spectrum optimization validating its robustness and suitability for high-throughput spectral correction pipelines.

A key design consideration in this work was the choice between a fully end-to-end deep learning model and the proposed DIRAS+ hybrid framework. While end-to-end architectures are highly expressive and can achieve excellent performance with sufficiently large and well-curated data sets, their application to Raman baseline correction is constrained

by several practical limitations. Such models require extensive, balanced data to generalize across diverse spectral profiles and may inadvertently encode chemically relevant features along with baseline trends, risking entanglement between background removal and analyte information. Their black-box nature also hinders interpretability and makes validation difficult in the absence of ground-truth baselines. Furthermore, their dependence on fixed input dimensions reduces adaptability to spectra acquired under varying resolutions or conditions, and their computational cost is often prohibitive for routine use.^{18,20}

In contrast, the DIRAS+ hybrid design combines machine learning–based λ prediction with the mathematically grounded DIRAS solver, offering both interpretability and efficiency. This structure allows intermediate steps to be examined and refined, provides robustness across data sets, and retains user control when corrections are needed. While preserving the automation advantage of end-to-end models, DIRAS+ reduces the learning burden by confining the model's role to identifying higher-level trends, with DIRAS ensuring precise and physically consistent baseline estimation.

DIRAS+ Excels on Real-World Data Sets and Enhances Downstream Analysis. To evaluate the performance of DIRAS+ using experimentally derived data, we applied ALS, SEALS, and DIRAS+ to two independent Raman data sets generated using SERS microsensors. These data sets are a suitable test set because they present a large range of S/N and background intensity. Spectra labeled AuTG-MPA were collected from SERS microsensors decorated with mercaptopropionic acid ($n = 296$) (Figure 8a) and spectra labeled AuTG-MPA-DAAQ are decorated with 2,6-diaminoanthraquinone ($n = 419$) (Figure 8g).

In general, the AuTG-MPA data set exhibited relatively uniform baseline, while the AuTG-MPA-DAAQ data set showed broader and more heterogeneous baselines, necessitating more adaptive regularization strategies. ALS and SEALS were run with optimized fixed λ values (10^4 and 10^6 , respectively), chosen to strike a balance between underfitting and over regularization, while DIRAS+ adaptively predicted λ for each spectrum. All methods were applied followed by consistent spectral processing: Savitzky-Golay filtering (window = 9, polynomial order = 2) and Euclidean normalization. In the baseline fitting examples (Figure 8b and 8h), ALS consistently underfit the global curvature, leaving behind low-frequency residuals. SEALS tended to overfit, particularly around broad or overlapping peaks. DIRAS+ avoided both failure modes, producing smooth corrections that tracked overall drift without distorting spectral features. These effects were also evident in the corrected spectra for the entire batch, shown in the average plots (Figure 8e and 8k). ALS-corrected spectra retained background elevation, while SEALS-corrected spectra displayed peak flattening. DIRAS+ preserved spectral integrity and curvature while minimizing baseline interference.

In the AuTG-MPA data set, ALS produced the highest batch AUC (27.65 ± 3.21), indicating baseline under subtraction across the group. SEALS yielded the lowest AUC (17.33 ± 1.45), suggesting over subtraction, while DIRAS+ achieved an intermediate and tighter distribution (18.96 ± 1.36), indicative of balanced correction across the batch and reflecting the appearance of the spectra (Figure 8b). These batch-level differences were statistically significant ($p < 0.001$ for ALS vs SEALS and SEALS vs DIRAS+ Figure 8f). The PCA of corrected spectra batches (Figure 8c) also showed that DIRAS

+ yielded the tightest cluster, reflecting reduced intersample variability and improved batch consistency. In the more complex AuTG-MPA-DAAQ data set, the baseline drift and curvature were more pronounced. Again, ALS failed to fully correct baseline drift (batch AUC = 24.98 ± 2.48), while SEALS overfitted (AUC = 18.75 ± 1.62). DIRAS+ achieved balanced correction (AUC = 19.70 ± 1.63) with lower intersample variance. Statistical tests confirmed significant batch-level differences across all comparisons ($p < 0.001$; Figure 8l). Despite SEALS showing a tight AUC distribution numerically, batch-level variability was apparent in both the corrected average $\pm$ std spectra (Figure 8j), where SEALS showed inflated standard deviation, and in PCA (Figure 8i), where SEALS-corrected spectra formed a broader, less cohesive cluster compared to DIRAS+.

We also evaluated the impact of baseline correction on quantitative signal fidelity, we applied SEALS and DIRAS+ to a downstream regression task using 3-mercaptopropionic acid (MPA) conjugated on gold nanostructures (AuTG-MPA). MPA acts as a label-free Raman reporter for local pH due to the well-characterized protonation deprotonation dynamics of its terminal carboxylic acid group. AuTG-MPA substrates were incubated in buffered solutions from pH 3 to 8 (1-unit intervals), and approximately 100 SERS spectra were acquired per condition. The COOH-to-COO[−] transition induces both peak intensity changes and minor positional shifts, particularly in the 904–932 cm^{−1} region. To isolate the influence of baseline correction on absolute intensity, we constructed a Boltzmann sigmoid regression model using the ratio of the pH-sensitive band at 904 cm^{−1} to a reference band at 1271 cm^{−1}. This low-complexity model is sensitive to baseline distortions and was used to evaluate preprocessing fidelity. The full data set was corrected using SEALS with three fixed λ values (10^4 , 10^5 , 10^6) and DIRAS+ with automated parameter prediction. Predicted pH values were recovered using the inverse Boltzmann fit.

SEALS at $\lambda = 10^5$ and 10^6 tuning conditions yielded R^2 values of 0.475 and 0.473, RMSEs of 1.219 and 1.250, and prediction accuracies within ± 0.5 pH of 44.3% and 50.9%, respectively (Figure S7). While SEALS at $\lambda = 10^4$ yielded an invalid negative R^2 (−1.049) due to overfitting. In contrast, DIRAS+ achieved markedly better performance without any manual parameter selection, with $R^2 = 0.832$, RMSE = 0.694, and ± 0.5 pH accuracy of 64.7% (Table 1). These results

Table 1. Calibration Curve Performance of SEALS (Fixed λ) vs DIRAS+

Method	R^2	RMSE (pH)	± 0.5 -unit accuracy %
SEALS 1e4	−1.049	2.522	33.483
SEALS 1e5	0.475	1.219	44.281
SEALS 1e6	0.473	1.250	50.862
DIRAS+	0.832	0.694	64.695

indicate that even moderate baseline distortions can significantly degrade downstream modeling accuracy in intensity-based tasks. The use of a Boltzmann model was intentional: it allowed us to isolate the effect of preprocessing on intensity fidelity without introducing model compensation. While ML algorithms can learn from peak curvature and shift even when baselines are distorted the Boltzmann model is strictly intensity-dependent and therefore acts as a direct diagnostic for baseline correction quality. Numerous Raman

studies rely on intensity, AUC or ratio-metric changes for analyte quantification and classification, including glucose detection, small molecule sensing, and cellular phenotype discrimination.^{30–33} In such systems, distortions in peak height can directly undermine interpretability and model output. Moreover, baseline correction challenges increase significantly in spectrally complex systems. For example, AuTG-MPA-DAAQ, which can be used in redox sensing, exhibit strong fluorescence and broad curvature, making them difficult to process with fixed-parameter methods. In such scenarios, DIRAS+ can provide a practical advantage by ensuring robust baseline correction without tuning. The improved accuracy observed in the MPA-based pH sensing task demonstrates the broader value of DIRAS+ and supports its role as a generalizable, tuneless preprocessing framework for high-throughput and intensity-critical Raman workflows.

DIRAS+ Offers Practical Runtime Suitability for High-Throughput Processing. Finally, we benchmarked SEALS, DIRAS, and DIRAS+ under identical preprocessing and hardware conditions to assess average runtime per spectrum. SEALS and DIRAS timings include a full baseline solve for each spectrum, while DIRAS+ timings incorporate feature extraction, autoencoder encoding, λ -prediction, and the final DIRAS solve. All one-time costs (data I/O, model loading) and cold-start bias were excluded. On our evaluation set, mean $\pm$ SD runtime per spectrum was 0.067 ± 0.036 s for SEALS, 0.036 ± 0.008 s for DIRAS, and 0.250 ± 0.020 s for DIRAS+ (Figure S8a). DIRAS is 46% faster than SEALS, whereas DIRAS+ is slower due to its added feature extraction and ML inference steps. A breakdown of DIRAS+ wall-time shows that the final DIRAS solve accounts for only 21% of total runtime, with the remainder distributed among feature extraction (18%), autoencoder encoding (31%), and λ -prediction (30%) (Figure S8b). While this introduces additional overhead, DIRAS+ provides fully automated λ -selection, eliminating manual tuning and enabling a reproducible, parameter-free baseline workflow, all while sustaining a practical throughput of ~ 5 spectra/s. Together, DIRAS and DIRAS+ offer complementary choices: DIRAS prioritizes speed, whereas DIRAS+ delivers full automation, allowing users to select the most suitable approach for their needs.

CONCLUSIONS

In this work, we have introduced DIRAS, a data-driven, adaptive baseline-correction framework that combines autoregressive spectral modeling with physics-inspired reweighting to deliver robust, flexible and interpretable Raman baseline removal in 36 ms per spectrum. Going beyond fixed-polynomial or generic smoothness-penalty methods, DIRAS employs a spectral energy driven reweighting strategy to update baseline weights dynamically, making it inherently less sensitive to the choice of smoothing parameter. By adapting to the underlying energy distribution of each spectrum, DIRAS maintains fidelity across a wide range of baseline profiles without extensive fine-tuning. For the ML-powered DIRAS+, we generated ground truth baselines using the 1-SSIM loss, which better preserves subtle peak shapes and structural features than traditional MSE. These high-quality targets enabled the predictor network to learn the optimal directly, yielding a truly parameter-free workflow that matches the performance of Optuna optimized baselines. Across synthetic benchmarks and SERS experiments, DIRAS+ consistently outperformed classical (ALS, arPLS) and recent (SEALS)

approaches in intensity fidelity, structural similarity and downstream predictive accuracy, even under challenging noise conditions and batch variability. By unifying a core mathematical model with machine-learning module, our pipeline not only reduces the burden of parameter-tuning but also provides a modular platform for future enhancements. DIRAS+ thus offers practitioners a practical, parameter-free solution for high-throughput and spectrally complex Raman applications empowering scientists to analyze intricate data sets with confidence, flexibility and minimal runtime overhead.

ASSOCIATED CONTENT

Data Availability Statement

The code and data used in this manuscript are available at https://github.com/Aradhya-sys/DIRAS_Plus.git

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.5c05185>.

Detailed descriptions of baseline correction algorithms (ALS, arPLS, SEALS, and DIRAS); sensitivity analysis of the regularization parameter (λ); comparative performance evaluation across varying SNR levels and batch data sets; Optuna-optimized DIRAS results; summary of optimization outcomes for spectra with increasing baseline complexity; ML model parity plots; inverse Boltzmann pH predictions; and mean processing time per spectrum (PDF)

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P.D.A.: Conceptualization, methodology, investigation, data curation, formal analysis, writing—original draft, writing—review and editing. S.M.: Investigation, writing—review and editing. R.D.G.: Formal supervisory role. C.J.C.: Formal supervisory role, experimental planning, writing—review and editing.

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Notes

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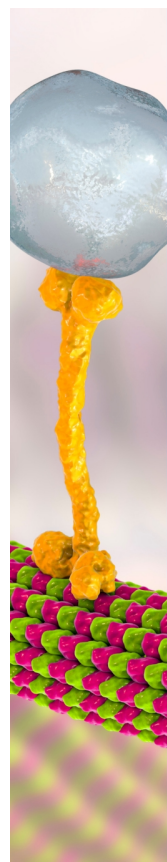
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ABBREVIATIONS

ALS, asymmetric least-squares; **arPLS**, asymmetrically reweighted penalized least-squares; **AuNP**, gold nanoparticle; **AuTG**, gold-coated Tentagel particles; **DAAQ**, 2,6-diaminoanthraquinone; **DIRAS**, dynamic iterative reweighted autoregressive spectral (baseline correction); **MPA**, 3-mercaptopropionic acid; **PLS**, penalized least-squares; **SEALS**, spectral estimation-based asymmetrically reweighted penalized least-squares; **SERS**, surface-enhanced Raman spectroscopy; **SSIM**, structural similarity index measure

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